Professorship of Energy Management Technologies
TUM School of Engineering and Design
Technical University of Munich

Energy Informatics Lab
Prof. Dr. Christoph Goebel

Assignment 6: Optimal Residential Energy System Sizing

# Meeting	Location	Content
1st Session	N3815	Introduction of assignment; work on training stage 1
Time until 2nd Meeting	Home	Work on training stage 1
2nd Session	N3815	Work on the competition stage 2; Competition; Presentations
Until one week after the 2nd Session	Home	Work on report

Task Description

The challenge of this assignment is to find a configuration of solar PV panels and batteries for a residential household such that the corresponding 20-year Net Present Value (NPV) is maximized. To be precise, you are supposed to find the following quantities:

- Peak power capacity a of PV module CAP_{PV} [kWp] and
- nominal energy capacity of a battery CAP_B [kWh].

We assume any non-negative capacity is admissible ($CAP_{PV} \geq 0$ and $CAP_B \geq 0$). You are given (resampled) data for one year (resolution: 1 min if not resampled, 12 h if resampled) of

- the household's expected electricity consumption $P_{HH,t}$ [kW] and
- weather data (ambient temperature $T_{a,t}$ [°C] and solar irradiance G_t [W/m^2]).

Photovoltaic Panels:

In addition, we will assume that the PV generation $P_{PV,t}$ [kW] behaves according to the following model:

$$P_{PV,t} = A_c \cdot G_t \cdot \eta_{PV,t} \cdot \frac{1}{1000},$$

where for the PV module area A_c [m^2], the PV module efficiency $\eta_{PV,t}$, and the PV cell temperature $T_{c,t}$ [°C] it holds that

$$A_c = \alpha^{-1} \cdot CAP_{PV},$$

$$\eta_{PV,t} = \eta_{ref} \cdot \left(1 - \beta \cdot \left(T_{c,t} - T_{c,ref}\right) + \gamma \cdot \log\left(\frac{G_t}{G_{ref}}\right)\right), \text{ if } G_t > 0, \text{ else } \eta_{PV,t} = 0,$$

$$T_{c,t} = T_{a,t} + h \cdot G_t.$$

Descriptions of the required constant parameters are given in Table 1. All data is given as CSV files, which are described in more detail at the bottom of this document.

Battery:

Further, we assume the following model for the battery power P_B [kW] (if the input data have a resolution of 1 min.):

$P_{B,t} = BatPower\left(SOC_{t-1}, P_{req,t}, CAP_B\right)$, if $CAP_B > 0$, else $P_{B,t} = 0$,

where

$BatPower\left(SOC_{t-1}, P_{req,t}, CAP_B\right) = -\min\left\{CAP_B \cdot 60\frac{1}{h},\ (1 - SOC_{t-1}) \cdot \frac{CAP_B \cdot 60\frac{1}{h}}{\eta_B}, -P_{req,t}\right\}$, if $P_{req,t} < 0$, else

$BatPower\left(SOC_{t-1}, P_{req,t}, CAP_B\right) = \min\left\{CAP_B \cdot 60\frac{1}{h}, SOC_{t-1} \cdot \eta_B \cdot CAP_B \cdot 60\frac{1}{h}, P_{req,t}\right\}$.

Moreover, the required power $P_{req,t}$ [kW] is given as

$$P_{req,t} = P_{HH,t} - P_{PV,t}.$$

Note that the battery power depends on the State of Charge SOC. This quantity changes in each time step (here: 1min), depending on the amount of energy fed into or drawn from the battery. It must be updated every time step via

$SOC_t = SOC_{t-1} - \frac{1}{\eta_B \cdot CAP_B \cdot 60\frac{1}{h}} \cdot P_{B,t}$, if $P_{B,t} > 0$, else

$SOC_t = SOC_{t-1} - \frac{\eta_B}{CAP_B \cdot 60\frac{1}{h}} \cdot P_{B,t}$.

As an initial value, we set $SOC_0 = 0$. In this setting, the battery power is positive, whenever we draw energy from the battery, and negative, whenever we feed energy into the battery.

Net Present Value Calculation:

For the NPV calculation, we need to know how much energy per time step has been drawn or fed into the grid. This amount $P_{D,t}$ is simply obtained via

$$P_{D,t} = P_{HH,t} - P_{PV,t} - P_{B,t}.$$

We are now ready to move on to the NPV calculations. For this, we will consider the three following aggregated quantities: $E_{HH,T}$, the summed up household load over one year T, $E_{D^+,T}$, the corresponding total energy drawn from the grid over one year T and $E_{D^-,T}$, the total excess solar energy fed into the grid over one year T.

We will assume the following costs:

- Initial PV investment $C_{0,PV} = c_{PV} \cdot CAP_{PV}$,

- Initial battery investment $C_{0,B} = c_B \cdot CAP_B$,
- Yearly PV maintenance costs $C_{M,PV} = c_{M,PV} \cdot C_{0,PV}$.

Moreover, we assume the following annual revenue:

$$R_T = E_{D^-,T} \cdot p_{FI} + (E_{HH,T} - E_{D^+,T}) \cdot p_{EL}$$

Assuming equal revenue each year, we then get the 20-year NPV via

$$NPV = -C_{0,PV} - C_{0,B} + \sum_{T=1}^{20} \frac{R_T - C_{M,PV}}{(1 + i_{INV})^T}.$$

Parameter	Definition
α	Peak power to area coefficient [kWp/m^2]
β	Solar PV module temperature coefficient of power [1/(°C)]
γ	Solar PV module solar irradiance coefficient of power
η_B	Battery efficiency
η_{ref}	Manufacturer PV module efficiency
c_B	Battery cost per kWh [€/kWh]
$c_{M,PV}$	Annual PV panel maintenance cost coefficient
c_{PV}	PV panel cost per kWp [€/kWp]
G_{ref}	Incident solar irradiance when η_{ref} is measured [W/m^2]
h	Ross coefficient [°C /(W/m^2)]
i_{INV}	Annual interest rate
p_{EL}	End-consumer electricity price [€/kWh]
p_{FI}	PV feed-in tariff [€/kWh]
$T_{c,ref}$	PV cell temperature when η_{ref} is measured [°C]

Table 1: Parameter descriptions and values.

Stage 1 – Training

Compute the optimal PV peak power and battery power capacity in the framework stated above. For this stage, you must download the following five files from Moodle:

- **DataA_params.csv**: Values corresponding to the parameters in Table 1,
- **hh_demand_12h_A.csv**: Mean 12-hour household load over one year,
- **hh_demand_true_A.csv**: Household load with one minute resolution over one year,

- **Munich_weather_12h.csv**: Mean 12-hour ambient temperature ('Ta') and solar irradiance ('G_Gk') over one year,
- **Munich_weather_1min.csv**: Three weeks (winter, spring, summer) of detailed ambient temperature ('Ta') and solar irradiance ('G_Gk') with one-minute resolution.

Household load data and weather data with a resolution of 12 h are provided for the whole year. Suppose data with a 1-minute resolution is only available for a few weeks, and data with a 12-hour resolution is available for the entire year. In that case, estimating the 1-minute data for the whole year is still possible (Stage 2).

In this stage, the household load data is also available for the entire year with a 1-minute resolution.

Stage 2 – Competition

Now, it is time to apply your code to new data for the competition. Please download the competition data from Moodle:

- **DataB_params.csv**: Values corresponding to the parameters in Table 1,
- **hh_demand_12h_B.csv**: Mean 12-hour household load over one year,
- **hh_demand_1min_B.csv**: Three weeks (winter, spring, summer) of detailed Household load with one-minute resolution.

Note that the weather data is the same as in stage 1.

Aim, Expected Outcome, and Grading

The aim of the assignment is to evaluate

- your understanding of how weather, consumption, and PV panel/battery characteristics jointly influence profit
- your ability to rigorously implement the above models
- your ability to find a way to optimize/approximate the optimal solution.

The expected outcome of the assignment includes

- a report of a maximum of two pages (excluding appendices),
- your results as a .json file.

The following aspects of your work will be graded:

- Formulation of physical models (correctness)
- Formulation of the optimization problem (correctness)
- Evaluation and interpretation of optimal solution
- Discussion (what went well, what went wrong, how the results could be improved–in your opinion, etc.)
- Competition ranking (just bonus points)

- General presentation of the assignment report (flow of information, quality of English, quality of illustrations, etc.)

This time, no evaluation notebook will be provided. We use the data with a minute resolution for evaluation. You can use the following value to make sure that you implemented the model correctly:

- In the training stage, for $CAP_{PV} = 1kWp$ and $CAP_B = 1kWh$, the NPV is approximately 337 €.

For the competition, please upload a .json file in the following form to Moodle:

{ `'name'`: your team's name as string,

`'cap_pv'`: float of optimal solar PV peak power capacity [kWp],

`'cap_b'`: float of optimal battery capacity [kWh],

`'npv'`: float of NPV [€]}.

Naming of Submissions

Each submission must be named according to the following pattern:

- Competition: f"Competition_6_Group_{Group_letter}.json“
- Presentation: f"Presentation_6_Group_{Group_letter}.pdf“
- Report and Code: f"Report_Code_6_Group_{Group_letter}.zip“

As an example for the group submissions of Group E:

- “Competition_6_Group_E.json”
- “Presentation_6_Group_E.pdf”
- "Report_Code_6_Group_E.zip“

Suggested Readings:

Goebel, C. & Cheng, V. & Jacobsen, H. (2017). Profitability of Residential Battery Energy Storage Combined with Solar Photovoltaics. Energies. 10. 976.

Notton, G. & Lazarov, V. & Stoyanov, L. (2010). Optimal sizing of a grid-connected PV system for various PV module technologies and inclinations, inverter efficiency characteristics and locations. Renewable Energy. 35. 2.

Energy Informatics Lab
Prof. Dr. Christoph Goebel

Appendix: Short description of data and various files

The files for this assignment provide all parameter values, required weather data, and household load.

DataA_params.csv

This file contains the parameter values for all parameters listed in Table 1 for the training stage.

DataB_params.csv

This file contains the parameter values for all parameters listed in Table 1 for the competition stage.

hh_demand_true_A.csv

This file contains the minute household load data for the whole year for the training stage, corresponding to P_{HH} [kW] in the model formulation stated above.

hh_demand_1min_B.csv

This file contains the detailed minute household load data for three weeks of the year (winter, spring, summer) for the competition stage, corresponding to P_{HH} [kW] in the model formulation stated above.

hh_demand_12h_A.csv

This file contains the mean 12-hour household load data [kW] for the training stage.

hh_demand_12h_B.csv

This file contains the mean 12-hour household load data [kW] for the competition stage.

Munich_weather_1min.csv

This file contains ambient temperature [°C] and solar irradiance [W/m^2] per minute in Munich for three weeks (winter, spring, summer) for the training and competition stage, created with Meteonorm[1], assuming a south-facing PV panel tilted by 30°.

Munich_weather_12h.csv

This file contains the mean ambient temperature [°C] and the mean solar irradiance [W/m^2] per 12 hours in Munich for the training and competition stage.

[1] Remund, J.; Kunz, S.; Shilter, C.; Müller, S. Meteonorm Version 6.0, Handbook; Meteotest: Bern, Switzerland, 2010.